

Teaching Observation Report

Arman Jahangiri – Math 397: Probability I

Exponential Distribution, Expected Value, Variance, and Memorylessness

TA observed	Arman Jahangiri
Course	Math 397: Probability I
Session type	Hybrid lecture
Topic	Exponential distribution, expected value, variance, memorylessness, and related probability concepts
Date	July 24, 2026

Overall Impression

This was an effective and carefully prepared hybrid lecture. Arman provided students with unusually strong organizational support through a well-structured course website, clear review materials, posted solutions, and neatly designed lecture notes. His explanations were calm, well paced, and well matched to the course level. Students received a coherent introduction to the exponential distribution that also reinforced earlier course ideas, including probability density functions and the geometric distribution.

The main opportunity for growth is not the mathematical clarity of the session, which was consistently strong, but the level and distribution of student participation. In a small class of about seven in-person students and several students on Zoom, there is considerable room to build a more interactive learning community.

Strengths and Pedagogical Impact

Arman began by situating the class within the broader course. His comments on the midterm, posted solutions, review sheet, and mini-project component communicated that the course has a clear structure and that students are being supported in multiple ways. The review sheet, in particular, gave students a useful way to identify the main ideas of the course and prepare purposefully. Making the midterm closely aligned with the practice and review materials also created a transparent assessment environment: students could focus their preparation on understanding the intended learning goals rather than guessing what might appear.

The course materials were a major strength. The organized GitHub course page and exceptionally neat notes reduce unnecessary organizational burden for students and make it easier to revisit ideas after class. Arman's summary slide at the end of the section and his use of learning goals when beginning new topics further helped students identify what mattered most.

Arman also made effective conceptual connections. When introducing the exponential distribution, he invited students to recall the geometric distribution and used the comparison to help distinguish the two. Similarly, he used new material as an opportunity to review how to verify that a function is a probability density function. These connections support durable learning because students are encouraged to organize probability concepts into a network of related ideas rather than treating each distribution as a separate procedure.

The Poll Everywhere question was another particularly effective choice. After working through one example, Arman gave students three minutes to apply a similar idea independently and reminded them to ask questions if they became stuck. The split between correct and incorrect responses made the debrief pedagogically valuable: instead of simply supplying the answer, Arman asked students to consider whether errors arose from algebra or from conceptual understanding. This kind of check gives both students and instructor useful information about what needs attention.

Arman's explanations of why expected value, variance, and medians matter were also valuable. Brief motivation through data matching and connections to later courses helped students see these quantities as tools with purposes, rather than formulas to memorize. During the proof of the memoryless property, he invited input on an idea students should already know. This was a good move because it asked students to retrieve and apply prior knowledge within a new argument.

Student Engagement and Learning Environment

The classroom atmosphere was calm, supportive, and professionally organized. Students asked thoughtful questions, including questions that extended from Arman's comments, which suggests genuine interest in the material. Students also appeared to follow parts of the memorylessness proof closely, as reflected in visible nodding.

At the same time, the overall participation pattern was fairly quiet. Much of the class appeared to be copying notes, and one student seemed to answer a disproportionate number of questions. A few students were also attending to phones or screens rather than the class at one point. These observations do not outweigh the strengths of the session, but they suggest that students would benefit from more regular invitations to process and contribute during the lecture.

Arman already has a strong starting point in the Poll Everywhere activity and in the questions he asks during proofs. Extending these practices could make student thinking more visible and distribute participation more widely. In such a small class, learning and using students' names would be especially helpful. Addressing students personally can strengthen connection, communicate that each student's contribution matters, and make it easier to invite a broader range of voices into the conversation.

Areas for Continued Growth

A useful next step would be to create more frequent, low-stakes moments for students to respond, predict, or briefly work before an explanation is completed. For example, during a proof, Arman might ask students to identify the next step, compare two distributions with a neighbor or through Zoom chat, or decide which earlier result is most relevant. These moments need not be lengthy. Their value is that they require students to retrieve and use ideas while the lecture is happening, rather than only record a finished solution.

A second refinement concerns pacing at familiar technical steps. Arman's careful detail was generally helpful, but in a Math 397 course some calculations, such as the mechanics of integration by parts in an expected-value computation, may not require the same amount of in-class attention. Keeping the full derivation available in the notes while moving more efficiently through standard algebraic steps could create time for a conceptual question, a brief student activity, or a discussion of interpretation.

Protocol Alignment

1. Classroom Organization & Clarity of Purpose

Protocol points addressed:

- Clear structure and pacing.
- Students understand the goal or purpose of the session.
- Logical flow of explanations or activities.

Evidence and interpretation. The session was organized around a clear progression of review, new material, examples, and a concluding summary. Learning goals and summary slides clarified the intended focus, while the lecture connected exponential distributions to earlier concepts and moved from explanation to individual application and debrief.

2. Communication & Instructional Delivery

Protocol points addressed:

- Clear explanations and mathematical communication.
- Effective use of board/visuals.
- Delivery students can follow, including pace, visibility, and audibility.

Evidence and interpretation. Arman's explanations were calm, appropriately paced, and well matched to the course level. The carefully organized notes, review sheet, course website, and summary slide gave students strong visual and organizational support. Comparisons, motivation, and explicit review helped students follow both new concepts and their relation to earlier material.

3. Student Engagement & Participation

Protocol points addressed:

- Students actively think, respond, or work.
- TA encourages participation and gives wait time.
- TA listens to and builds on student ideas.

Evidence and interpretation. Poll Everywhere gave students a meaningful opportunity to apply a similar problem independently, and students contributed answers and questions. Arman invited input during the lecture and asked students to diagnose the source of errors after the poll. More frequent low-stakes prompts could make engagement more visible and reduce reliance on a small number of frequent respondents.

4. Learning Environment & Rapport

Protocol points addressed:

- Welcoming, respectful tone.
- Encouragement of student questions.
- Constructive responses to confusion or errors.
- Approachability.
- Evidence of student focus, including any periods of distraction and possible causes.

Evidence and interpretation. The session had a calm, supportive tone, and Arman's course materials and assessment transparency communicated care for student success. Students asked questions beyond the immediate examples, indicating that questions were welcome. Students appeared attentive during important parts of the proof, though the overall room was quiet and a few students were off task. Learning and using names could strengthen connection and help invite participation from every student in this small hybrid class.

5. Support for Student Learning

Protocol points addressed:

- Checks for understanding.
- Addresses common difficulties.
- Connects the session to broader course goals.
- Students leave knowing the key idea of the day.

Evidence and interpretation. The Poll Everywhere question gave Arman clear evidence of students' understanding and misunderstanding. Asking whether errors were algebraic or conceptual was a productive way to identify the kind of support students needed. Arman connected distributions to one another and motivated expected value, variance, and medians through their uses and future relevance. The learning goals, repeated conceptual connections, and final summary supported students in identifying the central takeaways.

6. Group Work Organization: Not Applicable

Formal group work was not part of the observed session.

Summary

Arman gave a strong, well-prepared hybrid lecture. His clear explanations, excellent course materials, purposeful review, and thoughtful use of Poll Everywhere supported students' conceptual understanding of the exponential distribution and its connections to earlier probability topics. The principal refinement is to build more frequent opportunities for students to contribute and to use the small class size to create a more personal, participatory learning community.